

Open Data Portal Port Management



The Team



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05

Personas and Scenarios

Users and their interaction

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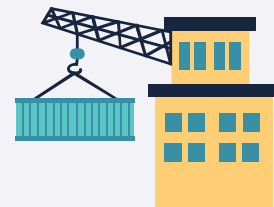
System Diagrams

The system's data structure and architectural view

07

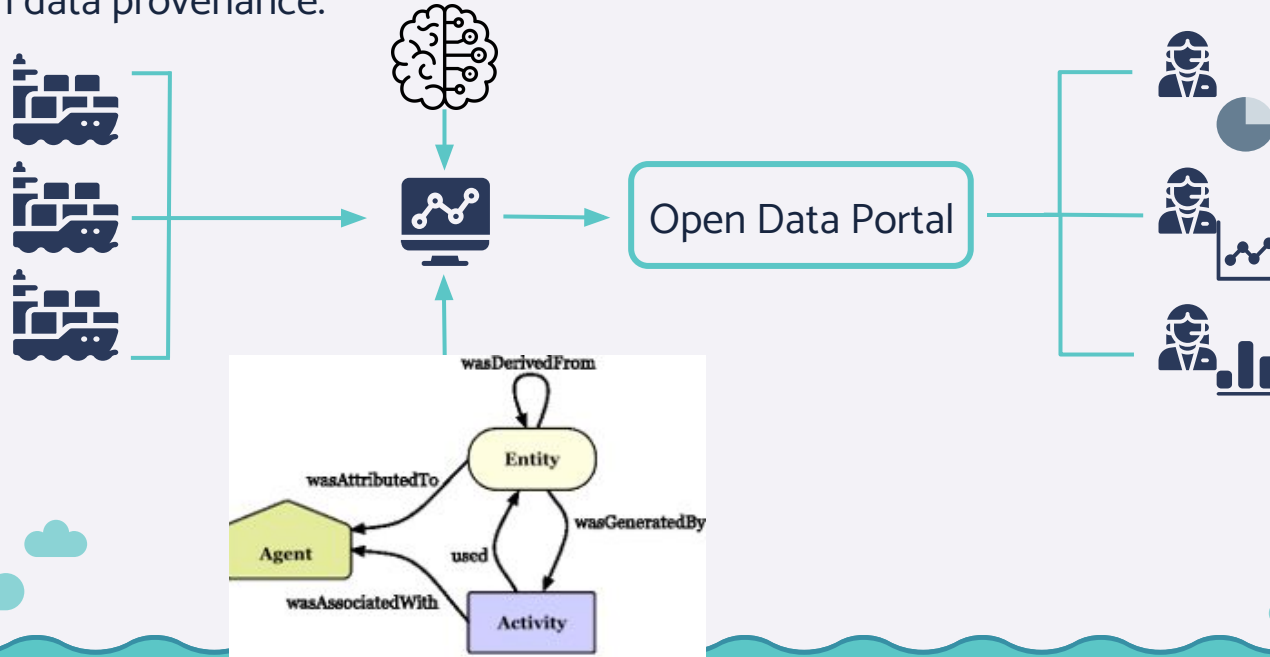
Organization

Project's current state and future iterations



01 Context

The need for a single portal for port management data analytics with data provenance.



01

State of the Art



OpenPort ¹



Softship ²



Piveau ³



Open Data Portal ⁴

Data Storage	✓	✓	✓	✓
Analytics	✓	✓	✓	✓
Maritime Data	✓	✓	✗	✓
Data Provenance	✗	✗	✓	✓
Data completeness with ML model	✗	✗	✗	✓
Portuguese	✓	✗	✗	✓

1-<https://openport.com.br/>

2-<https://www.softship.com/>

3-<https://www.piveau.de/en/>

4-<https://pi-open-data-portal.github.io/>

02



Functional Requirements

Administrator should be able to:

- Manage User accounts
- View User interaction with the system
- Manage Data quality

User should be able to:

- Login/register
- See the data and analytics
- Interact with and filter the evaluation metrics
- Download and export data
- Use ML algorithm to complete missing fields
- Use Text processing to extract data from descriptions
- Add Provenance to Data
- Upload Data to the system



02



Non-functional Requirements



Performance



Data Protection
and Governance



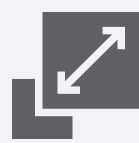
Availability



Usability



Maintainability

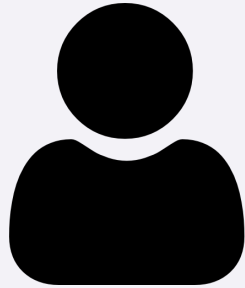


Scalability



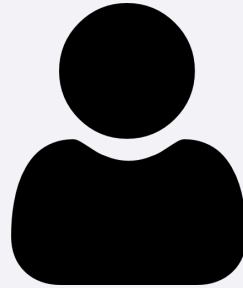
03

Actors



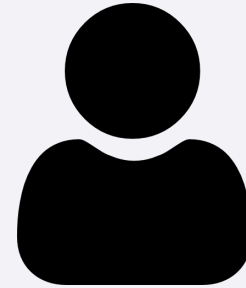
Administrator

Responsible for the maintenance, security, and efficiency of the Open Data Portal



Data Scientist

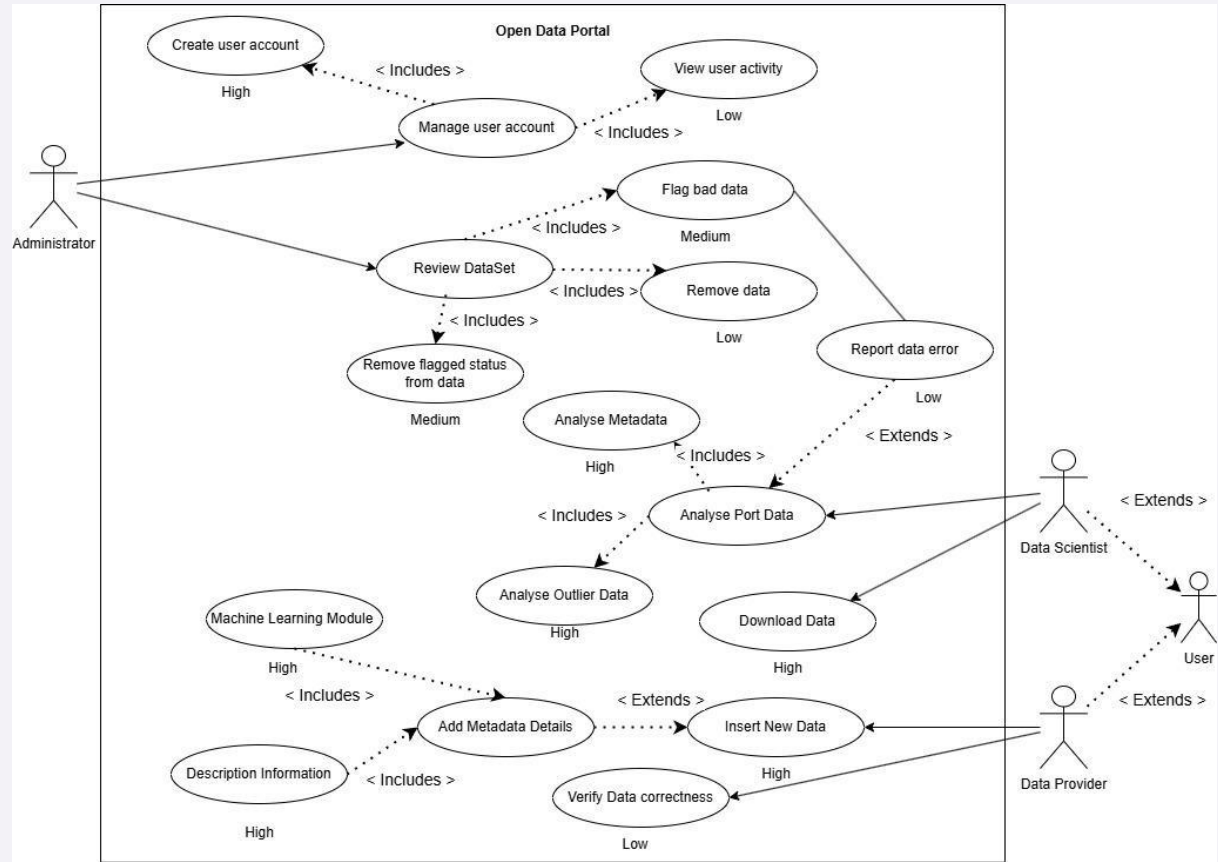
Responsible for collecting, analyzing, and interpreting large volumes of data.



Data Provider

Responsible for collecting, curating, and publishing maritime and logistics data on the Open Data Portal.

04 Use Cases



05

Personas



Name: Carlos Nogueira

Age: 42 years old

Sex: Male

Profession: DB engineer



05



Scenario

Carlos Nogueira, the IT administrator, wants to create a new account for a user, so that he can interact with the open portal data system.



Carlos logs into the system and goes to the user management section



Fills out a form to add information about the new user



Defines with permissions the user was



Sends a email through the system to the user.



The user activates the new account



Carlos can view the status of the user



05

Mockups



Admin Page

User Management

Search...

Viewers - Open Data Portal

Name	Level	Account Created Date	Role Created Date	Actions
 Matthew Wilson	Data Provider	06 -03-2021	06 -03-2021	 
 Sarah Martinez	Admin	06 -03-2021	06 -03-2021	 
 Christopher Brown	Data Provider	18 -03-2021	20 -03-2021	 
 Emily Thompson	Data Provider	11 -06-2021	11 -06-2021	 
 David Smith	Data Provider	25 -06-2021	28 -06-2021	 



Add User

Data reports



 Data Error
Table Ships Row 2

See error



 Data Error
Table Ships Row 204

See error



 Data Error
Table Containers Row 104

See error





05

Personas



Name: Mariana Costa

Age: 39 years old

Sex: Female

Profession: Senior Data Scientist



05



Scenario

As a Data Scientist, Mariana needs to analyze data to be able to improve the efficiency of logistics between various ports.



Mariana logs into the portal



Navigate through the dashboard that contains the various datasets available.



Select the data for the port she wants.



Select which elements of the dataset she needs.



Limit the time frame

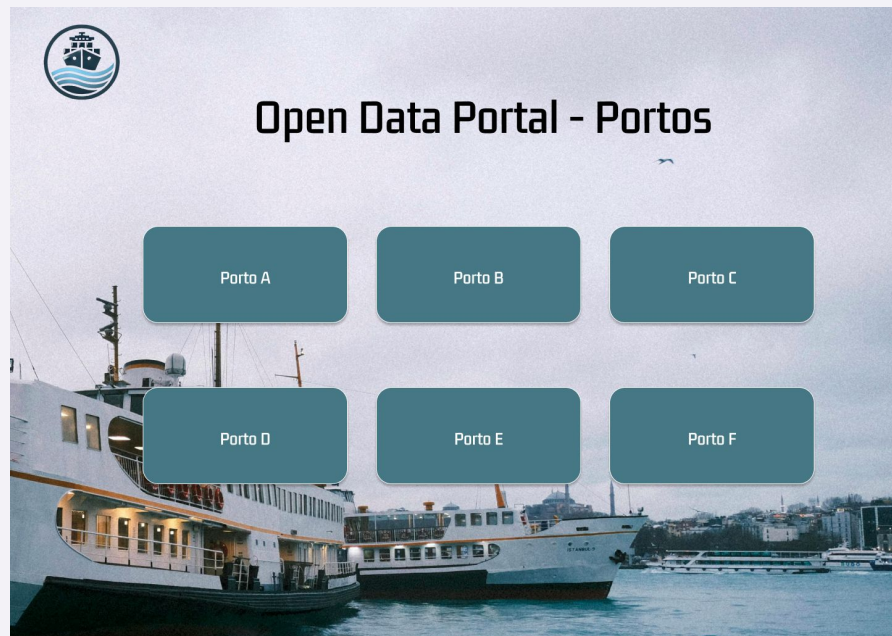
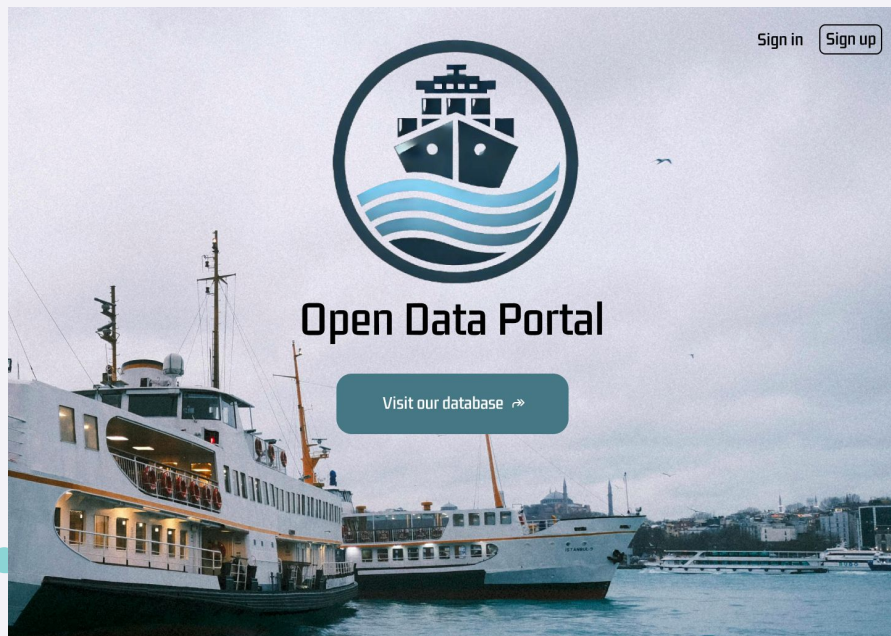


Analyse the data



05

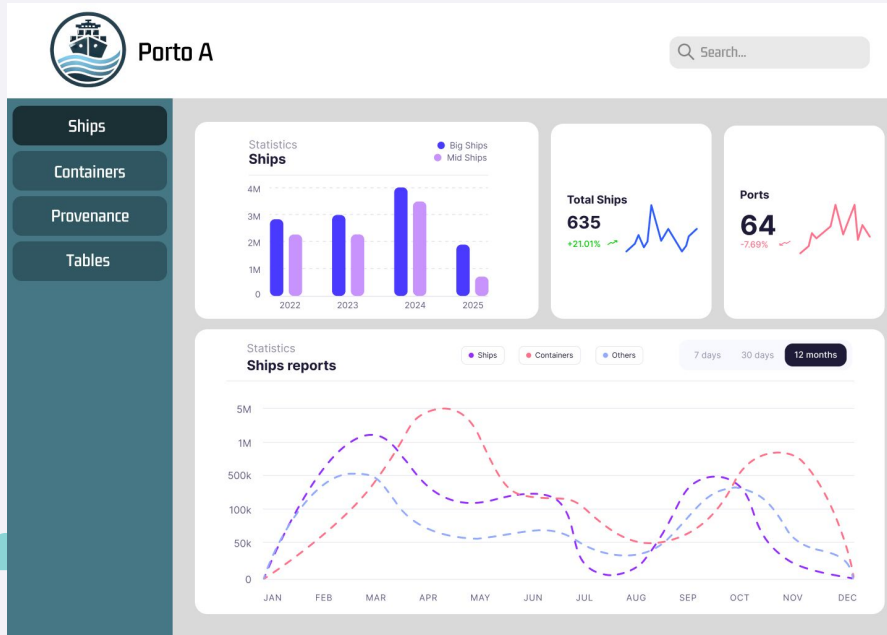
Mockups





05

Mockups



Statistics
Ships reports

Ships

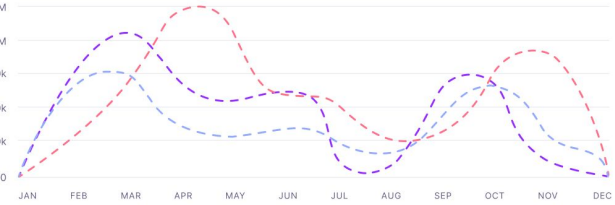
Containers

Others

7 days

30 days

12 months



05

Personas



Name: João Silva

Age: 32 years old

Sex: Male

Profession: Port Worker

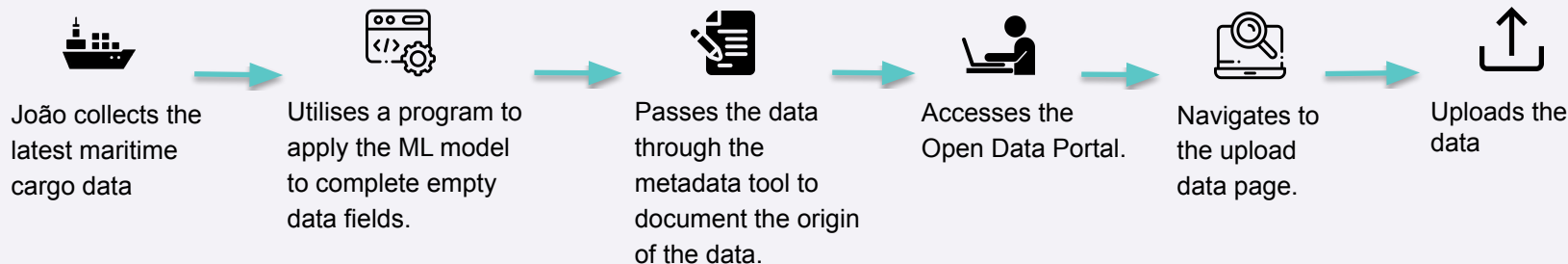


05

Scenario



As a Port Worker, João needs to be able to upload new data on a ship's cargo.



05

Mockups



Porto A

Search...

Download data

Ships

Containers

Provenance

Tables

Table Ships

Table Stock

Table Provenance

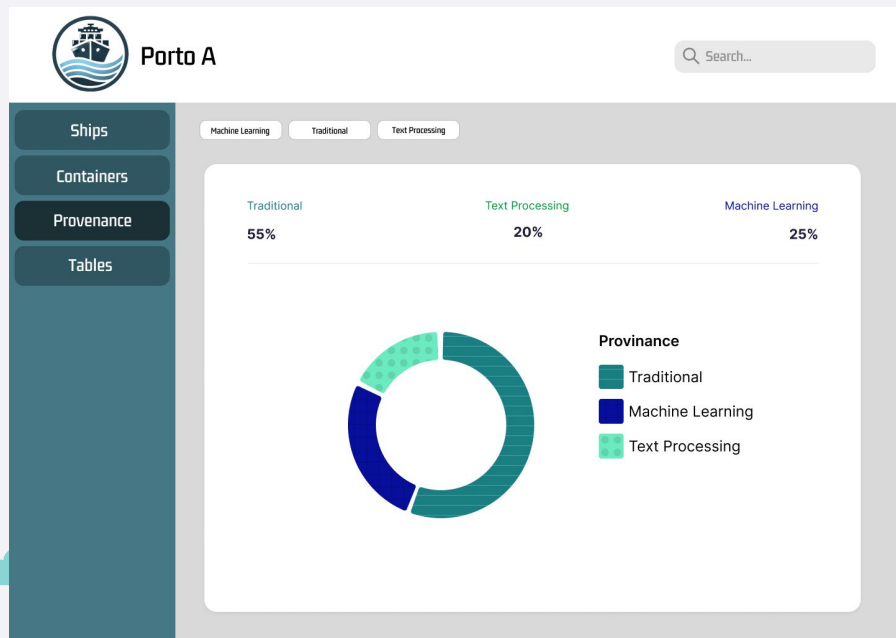
ID	NAME	DESCRIPTION	STATUS	DATE	ORIGEM	DEPOSIT
1	Ship 1 5684236526	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	04/02/25 12 May	Lisboa LIBS	Porto OPOR
2	Ship 2 5684236527	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	On	02/03/25 12 Apr	Porto OPOR	Aveiro AVR
3	Ship 3 5684236528	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	01/01/24 15 Oct	Aveiro AVR	Lisboa LIBS
4	Ship 4 5684236529	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Inactive	09/11/24 20 Jul	Algarve ALG	Porto OPOR
5	Ship 5 5684236530	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	12/04/24 1 Jan	Porto OPOR	Lisboa LIBS
6	Ship 6 5684236531	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	On	23/07/24 2 Feb	Porto OPOR	Aveiro AVR
7	Ship 7 5684236532	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Due	21/05/24 3 Mar	Lisboa LIBS	Porto OPOR
8	Ship 8 5684236533	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	12/03/24 14 Sep	Lisboa LIBS	Porto OPOR
9	Ship 9 5684236534	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Inactive	01/01/24 14 Dec	Porto OPOR	Lisboa LIBS

1-10 of 97

Rows per page: 9 < 1/10 >

05

Others Mockups



 Porto A

Q Search...

Machine Learning Traditional Text Processing

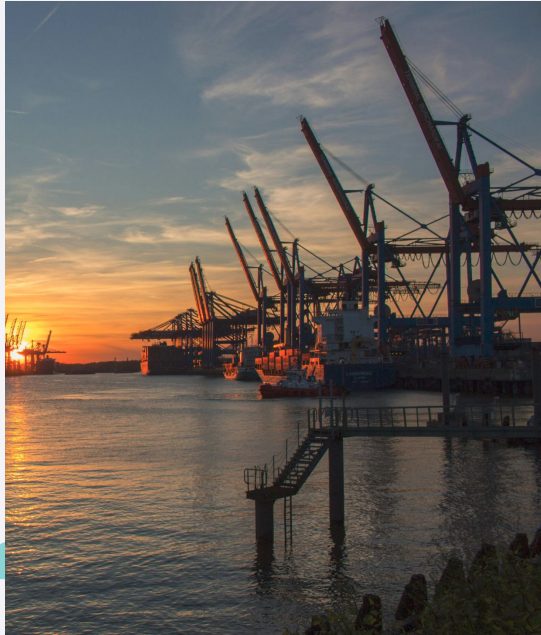
Machine Learning Rows [Download data](#)

Name	Origem - Destino	Ship			Description
		Type	Id	Weight	
Ship 1	Santos - Rotterdam	Big	1	5000	Transporte de eletrônicos
Ship 2	Miami - Lisboa	Small	2	12000	Carga de móveis e utensílios domésticos
Ship 3	Shangai - Los Angeles	Big	3	25000	Transporte de petróleo bruto
Ship 4	Porto - Milão	Mid	4	27000	Envio de veículos automotores, incluindo carros de luxo e caminhões.
Ship 5	Cabo Verde - Lisboa	Mid	5	11000	Envio de produtos químicos industriais altamente regulamentados.
Ship 6	Lisboa - Porto	Small	6	9999	Transporte de maquinário pesado para indústrias de mineração e construção.
Ship 7	Porto - Lisboa	Big	7	7777	Transporte de petróleo bruto para refinarias na Europa.



05

Others Mockups



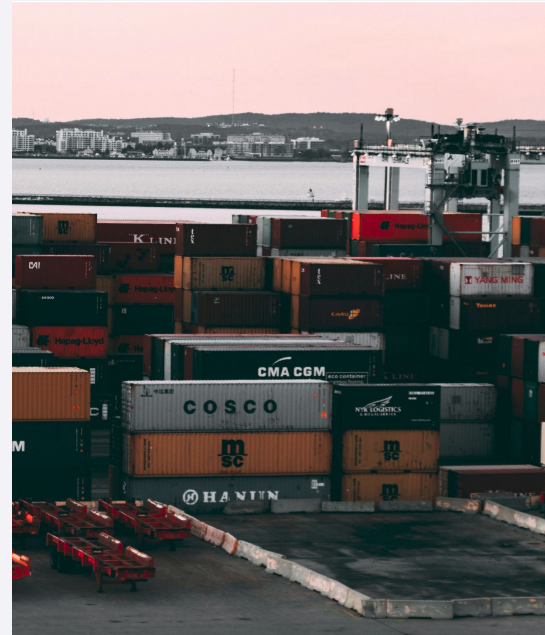
Hello Again!
Welcome Back

 Email Address

 Password

Login

[Forgot Password](#)



Hello!
Sign Up to Get Started

 Full Name

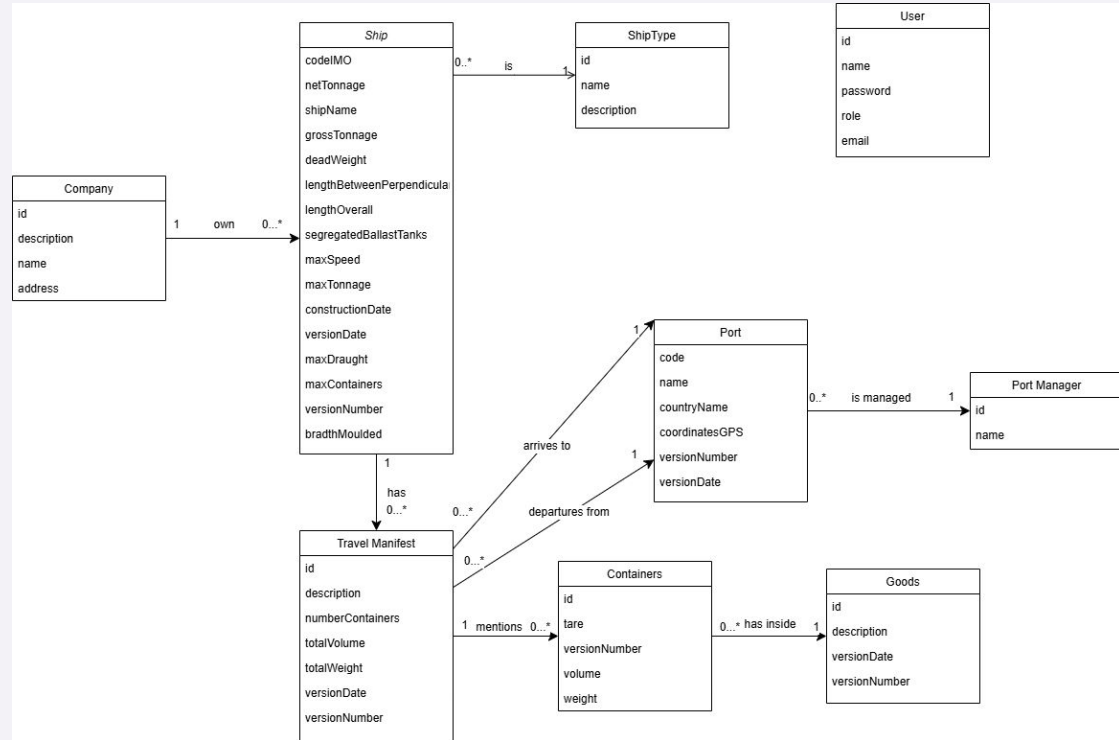
 Email Address

 Password

Register

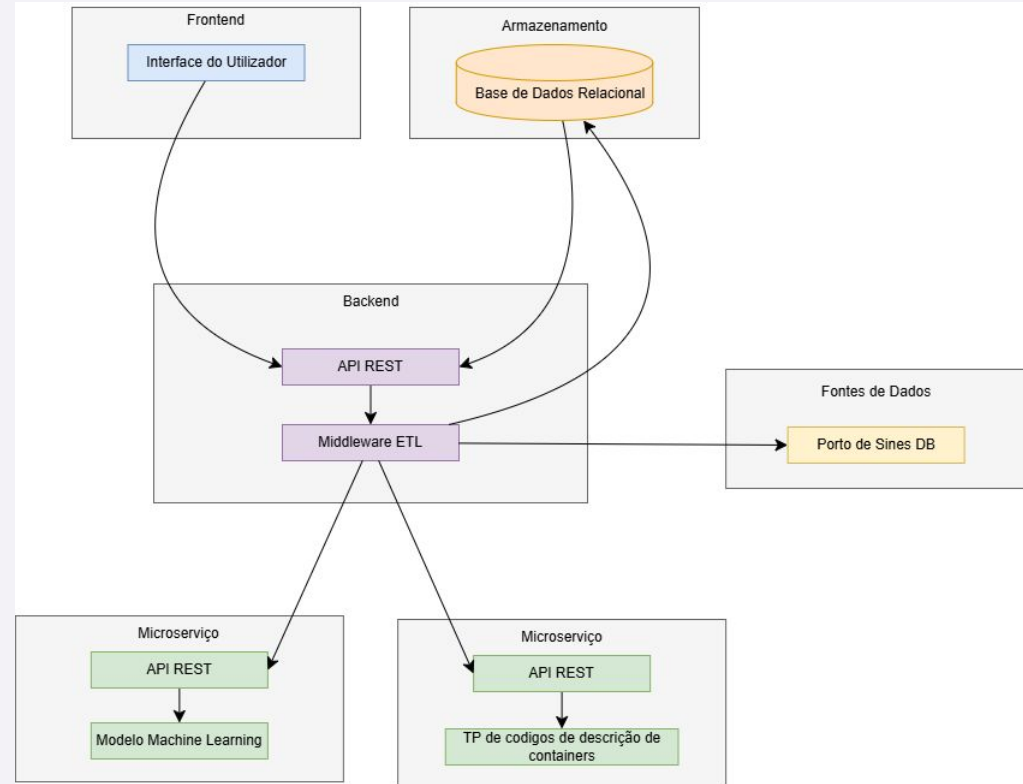
06

Domain Diagram



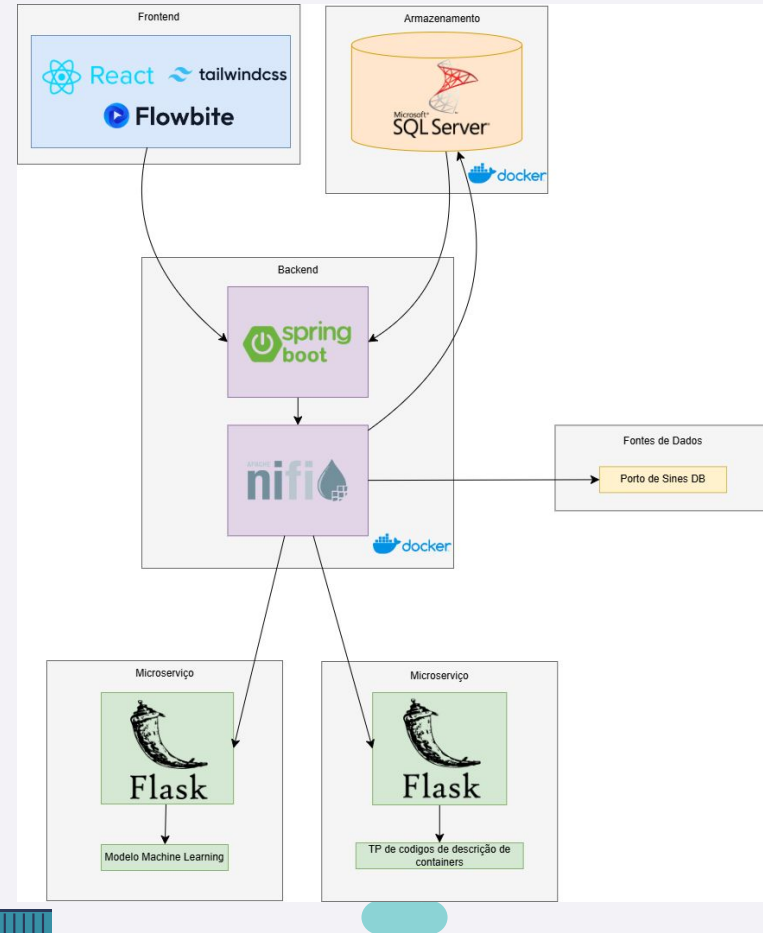
06

Architecture Diagram



06

Architecture Tech Diagram



07

Organization

The screenshot displays a Kanban board interface with the following columns and tasks:

- No Status (10):**
 - PL_Project #21 [UserStory] Create user account
 - PL_Project #22 [UserStory] Download Port data
 - PL_Project #23 [UserStory] Analyse Port data
 - PL_Project #24 [UserStory] Manage user accounts
 - PL_Project #25 [UserStory] Flag bad data
 - PL_Project #26 [UserStory] Remove Data
 - PL_Project #27 [UserStory] Remove Flagged status
 - PL_Project #28 [UserStory] Insert new data
 - PL_Project #29 [UserStory] Analyse metadata
 - PL_Project #30 [UserStory] Analyse outlier
- Todo (2):**
 - PL_Project #14 [Documentation] Create DataSet visualization module in the prototype in figma
 - PL_Project #19 [Documentation] Update WebSite with new information
- In Progress (5):**
 - PL_Project #11 [Documentation] Develop a prototype of the project
 - PL_Project #12 [Documentation] Create Initial WebPage of the prototype in figma
 - PL_Project #16 [Documentation] Define the Personas of the project
 - PL_Project #15 [Documentation] Add Data provenance components in the prototype in figma
 - PL_Project #32 [Documentation] Create User Management module in the prototype in figma
- In Review (1):**
 - PL_Project #10 [Documentation] Creation of M2(Elaboration) powerpoint
- Testing (0):**
- Done (11):**
 - PL_Project #6 [Feature] Create mini-Website of project
 - PL_Project #13 [Documentation] Create Analyses module in the prototype in figma
 - PL_Project #17 [Documentation] Create some Scenario in the project
 - PL_Project #7 [Feature] Deployment of Mini-website
 - PL_Project #8 [Organization] Implement initial stture of project in github
 - PL_Project #9 [Documentation] Creation of M1(ception) powerpoint
 - PL_Project #3 [Documentation] Architecture Design and Structure



07



Calendar

Steps	Week							
	1	2	3	4	5	6	7	8
Inception								
Definition of technologies to be used								
Uses cases								
Architecture design								
Explore the original database								
Creation of the database with metadata								
Implementation of the API								
Development of the website UI								



07



Calendar

Week

Steps	9	10	11	12	13	14	15	16	17
Development of the website UI									
Data prediction and outlier detection									
Prototype									
Metadata visualization									
Analysis module									
Finalization of the website UI									
Deployment									
Final Report									



Thanks!



Open Data Portal
Port Management

